

Probabilities

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What are probabilities?

- **Probability as frequency of outcome of events:**
 - In this way of thinking about probabilities we look at the number of times a given event happens over a large number of tries.
 - It is difficult to define consistently however, without running into circular reasoning.
- **Probability as degree of belief:**
 - Subjective probability is associated with personal judgements about how likely something is to happen.
 - For example, 'I believe that team X will beat team Y, because team Y's star player has an injury, while team X has been training really hard.' Such statements can be made even if teams X and Y have never played each other.

- By requiring that two people arrive at the same conclusion if given the same assumptions and data, this definition of probability can be formalised into a mathematical system equivalent to the other definitions.
- **Probably derived from axioms:**
 - Probability is a measure that satisfies a set of axioms derived from logic and set theory, such as the Kolmogorov or Cox axioms.
 - This sidesteps the frequentist vs Bayesian interpretation by sticking to purely mathematical concepts.

In this course we start out with this definition. In general we will follow the Bayesian degree-of-belief way of thinking about probability.

Notation and basic concepts

Set notation

- A set is a collection of elements, e.g.: $A = \{1, 2, 3\}$
- $e \in A$ means e is a member of the set A , e.g.: $1 \in \{1, 2, 3\}$
- A set can also be represented by a rule: $\{x | x \text{ follows a rule}\}$

For example, the set E of even integers: $E = \{x | x = 2y, y \in \mathbb{Z}\}$

- Set inclusion ($\subseteq$). A is included in B (or is a subset of B) if all the elements of A are also elements of B .

For example: $\{1, 2\} \subseteq \{1, 2, 3\}$

Set operators

Let $A = \{1, 3, 5\}$; $B = \{2, 3, 4\}$

- **Union** $\cup$ All elements of A and all elements of B

$$A \cup B = \{1, 2, 3, 4, 5\}$$

- **Intersection** $\cap$: Elements that are in both A and B

$$A \cap B = \{3\}$$

- **Difference** $\setminus$ Elements that are in A but not in B

$$A \setminus B = \{1, 5\}$$

$$B \setminus A = \{2, 4\}$$

- **Complement:**

The complement of A in reference to Ω includes all elements in Ω that are not in A. For the die example $\Omega = \{1, 2, 3, 4, 5, 6\}$,

$$A^c = \{2, 4, 6\} \text{ or}$$

$$A^c = \{\omega : \omega \in \Omega \text{ and } \omega \notin A\}$$

- **Empty set** $\emptyset$

The empty set, $\emptyset$, is the complement of the universal set:

$$\Omega^c = \emptyset \text{ and } \emptyset^c = \Omega.$$

This means, $A \cup \emptyset = A$ and $A \cap \emptyset = \emptyset$.

- **Power set**

Collection of all possible subsets of a given set

$$A = \{1, 3, 5\}$$

$$\mathcal{P}(A) = \{\emptyset, \{1\}, \{3\}, \{5\}, \{1, 3\}, \{1, 5\}, \{3, 5\}, \{1, 3, 5\}\}$$

Outcomes, events, probability

Outcomes and sample space

The outcomes ω of an experiment are elements of the set of all possible outcomes, called the sample space Ω .

Consider the experiment of tossing a (fair) coin twice:

- $\omega = \text{HH}$ ("two heads")
- $\omega \in \Omega = \{\text{HH}, \text{HT}, \text{TH}, \text{TT}\}$

Events and event space

An event F is a set of outcomes

- $F = \{\text{HH}, \text{HT}, \text{TH}\}$ ("at least one head")

Events are elements of the event space $\mathcal{F}$: the power set of the sample space (the set of all possible outcomes)

Note that Ω and $\mathcal{F}$ are not the same. The sample space contains the basic outcomes and event space contains sets of outcomes.

Probability

The probability function P assigns a probability (a number between 0 and 1) to events

- $F = \{HH, HT, TH\}$
- $\Pr(F) = \frac{3}{4}$

Kolmogorov's axioms of probability

- The probability measure of events is a real number equal or larger than 0:

$$0 \leq \Pr(A)$$

- The probability measure of the universal set is 1.

$$\Pr(\Omega) = 1$$

- If the sets $A_1, A_2, A_3 \dots \in \mathcal{F}$ are disjoint, then

$$\Pr(A_1 \cup A_2 \cup \dots) = \Pr(A_1) + \Pr(A_2) + \dots$$

Consequences of the axioms of probability

- Numeric bound:

$$0 \leq \Pr(A) \leq 1$$

- Monotonicity:

$$A \subseteq B \text{ then } \Pr(A) \leq \Pr(B)$$

- Complement rule:

$$\Pr(A^c) = 1 - \Pr(A)$$

- Sum rule:

$$\Pr(A \cup B) = \Pr(A) + \Pr(B) - \Pr(A \cap B)$$

Example, a single fair die:

- $\Omega = \{1, 2, 3, 4, 5, 6\}$
- $\Pr(\omega) = \frac{1}{6} \quad \forall \omega \in \Omega$

- Events $A = \{1, 3\}$ and $B = \{1, 2, 3, 4\}$

- Monotonicity:

$$A \subseteq B, \Pr(A) = \frac{1}{3} \leq \Pr(B) = \frac{2}{3}$$

- Complement rule:

$$\Pr(A^c) = \Pr(\{2, 4, 5, 6\}) = \frac{2}{3} = 1 - \Pr(A)$$

- Sum rule:

$$\Pr(A \cup B) = \Pr(\{1, 2, 3, 4\}) = \frac{2}{3}$$

$$\Pr(A \cup B) = \Pr(A) + \Pr(B) - \Pr(A \cap B) = \frac{1}{3} + \frac{2}{3} - \Pr(\{1, 3\}) = \frac{2}{3}$$

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Two fair six-sided dice are rolled. What's the probability to obtain two sixes?

- 1/25
- 1/36
- 2/36
- 1/6

Conditional probabilities and independence

Conditional probabilities

The conditional probability of event A happening, given that event B happened, is

$$\Pr(A|B) = \frac{\Pr(A \cap B)}{\Pr(B)}$$

Instead of the Kolmogorov axioms, probability theory can also be defined in terms of conditional probabilities, using the Cox axioms.

Independence

If A is independent of B, $\Pr(A|B) = \Pr(A)$: the conditional probability of A given B does not depend on B. From this follows that

$$\Pr(A \cap B) = \Pr(A) \Pr(B)$$

Law of total probability

Let $\{H_1, H_2, \dots\}$ be a countable collection of sets which is a partition of Ω , where

$$H_i \cap H_j = \emptyset \text{ for } i \neq j$$

$$H_1 \cup H_2 \cup \dots = \Omega$$

The probability of an event D can be calculated as

$$\Pr(D) = \Pr(D \cap H_1) + \Pr(D \cap H_2) + \dots$$

or in terms of conditional probabilities

$$\Pr(D) = \Pr(D|H_1) \Pr(H_1) + \Pr(D|H_2) \Pr(H_2) + \dots$$

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Two fair six-sided dice are rolled. Let A be the event that the sum of the dice is 7, and let B be the event that at least one of the two dice is a 4. Are A and B independent?



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What about when B is the event that the first die is a 4?



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Bayes' theorem

Applying the definition of the conditional probability twice we get Bayes' theorem:

$$\Pr(A|B) = \frac{\Pr(A \cap B)}{\Pr(B)} = \frac{\Pr(B|A) \Pr(A)}{\Pr(B)}$$

Named after Thomas Bayes, British clergyman, 1702-1761

Clicker: a test for rare events

Let us assume there is a rare disease that affects 0.1% of the population. There is a test that can detect this disease. It has a detection efficiency of 99% and a probability of error (false-positive) of 2%.

What is the probability $\Pr(D|+)$ of having the disease when receiving a positive test?

$$\Pr(D|+) = \frac{\Pr(+|D) \Pr(D)}{\Pr(+)} = \frac{\Pr(+|D) \Pr(D)}{\Pr(+|D) \Pr(D) + \Pr(+|D^c) \Pr(D^c)} \text{ (law of total probability)}$$

- $\Pr(+|D) = 0.99$: probability of a positive test result, given the disease is present (detection efficiency of 99%)

- $\Pr(D) = 0.001$: the disease affects 0.1% of the population
- $\Pr(+|D^c) = 0.02$: probability of a positive test result, given the disease is not present (false-positive rate of 2%)

$$\Pr(D|+) = \frac{0.99 \cdot 0.001}{0.99 \cdot 0.001 + 0.02 \cdot 0.999} = 0.047$$

The disease is only present in 5% of the cases where the test is positive!